

Yuhui Wang

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SUMMARY

PhD student in Computer Science, Stony Brook University, USA. Specializing in LLM safety & trustworthiness. 3 First-authored paper in ICLR and IEEE S&P, \$5K fellowship awardee. Available for Summer 2026 (May to Sept) internship. Expt Grad in 2029.

EDUCATION

Stony Brook University Computer Science PhD Student NY, USA	2024.09 - Present
• Advisor: Prof. Ting Wang • Research Interest: LLM Safety Alignment and Trustworthiness, Security of AI-based Systems	
Shanghai Jiao Tong University Industrial Engineering Master's Degree Shanghai, China	2021.09 - 2024.03
• Advisor: Prof. Di Wang • Research Interest: Multi-dimensional Time-series Classification & Prediction	
Zhejiang University Industrial Engineering Bachelor's Degree Hangzhou, China	2017.09 - 2021.06

RESEARCH EXPERIENCE

LLM Safety Alignment & Trustworthiness Project Leader 2 First-Authored ICLR Papers	2024.09 - Present
• Found the vulnerabilities of existing safeguards against intensive fine-tuning jailbreak attacks. Therefore, we proposed a self-destructive defense with a custom loss coupling benign and harmful gradients, enabling the model to retain ~5% initial harmful metric or be fully destroyed post-attack; implemented via adapting HuggingFace Trainer	
• Proposed a perturbation framework to prove and quantify the CoT and memory's joint impacts on the final answer. Based on this finding, I suppressed retrieval shortcuts in reinforcement learning to strengthen model reasoning and ensure trustworthy outputs; improved CoT robustness by 47.8% and pass@1 accuracy by 5.8% across diverse benchmarks, implemented through customized adaptation of the veRL trainer	

GraphRAG Data Poisoning Project Leader 1 Cofirst-Authored IEEE S&P Paper	2024.09 - 2025.02
• As the first data poisoning study on GraphRAG developed by Microsoft, we proposed GragPoison, a poisoning attack targeting GraphRAG by leveraging its unique relation and entity retrieval mechanism to amplify the poisoning effect . Achieved up to 98% attack success with <68% poisoning text on medical, cybersecurity, and commonsense datasets across multiple GraphRAG variants compared with benchmarks	

Adaptive LLM & Agent Jailbreak & Defense 1 ICLR Paper, 3 Papers Under Review at ICML	2024.09 - Present
• Proposed an adaptive long-horizon prompt injection attack that iteratively refines injected observations based on agent tool-call trajectories, enabling covert bridging between benign and malicious actions; evaluated on AgentDojo environments, achieving up to 93.1% attack success rate.	
• Participated in developing AutoRAN, an automated jailbreak attack framework that uses a red-team reasoning model to simulate and refine prompts via target models' intermediate reasoning steps. Achieved ~100% success rates on GPT-o3/o4-mini and Gemini-2.5-Flash	
• Participated in developing RASA, a routing-aware expert-level safety alignment framework for MoE LLMs that eliminates routing shortcuts, achieving near-perfect jailbreak robustness (~ 100% harmlessness across mixed attacks) while preserving general performance on MMLU, GSM8K, and TruthfulQA with consistent over-refusal performance	

SKILLS

• Programming: Python, SQL, C++, Java, C	
• ML/AI Frameworks: PyTorch, HuggingFace, vLLM, GraphRAG, veRL, Ray, Megatron, FSDP, DeepSpeed, wandb, scikit-learn, rllm	
• Specialized Skills: LLM Post-Training (LoRA, RLHF, DPO, PPO, GRPO, DAPO, GSPO), Adversarial ML, RAG, Prompt Engineering, Jailbreak Defenses, Model Steering, LLM Agent, Prompt Injection, Machine Learning, Software Engineering, Agentic RL	

AWARDS & SERVICE

Nisha and Vinod K. Singhi Graduate Fellowship (1 PhD student/year in CS Dept., \$5,000), Stony Brook University	2024, 2025
Reviewer of NeurIPS 2025, ICLR 2026, IEEE CASE 2025&2024, IEEE TASE	2024, 2025
Teaching Assistant of CSE352 & CSE582 — Office hours, Grading, Student Communication	2024, 2025

SELECTED PUBLICATIONS

Google Scholar: scholar.google.com/citations?user=eJgbw-oAAAAJ&hl=en

[1] **Yuhui Wang**, Changjiang Li, Guangke Chen, Jiacheng Liang, Ting Wang, "Reasoning or Retrieval? A Study of Answer Attribution on Large Reasoning Models", International Conference on Learning Representations (**ICLR**), 2026

[2] **Yuhui Wang**, Rongyi Zhu, Ting Wang, "Self-Destructive Language Model", International Conference on Learning Representations (**ICLR**), 2026

[3] Jiacheng Liang*, **Yuhui Wang***, Changjiang Li, Rongyi Zhu, Tanqiu Jiang, Neil Gong, Ting Wang, "GraphRAG under Fire", IEEE Symposium on Security and Privacy (**IEEE S&P**), 2026

[4] Tanqiu Jiang, Zian Wang, Jiacheng Liang, Changjiang Li, **Yuhui Wang**, Ting Wang, "RobustKV: Defending Large Language Models against Jailbreak Attacks via KV Eviction", International Conference on Learning Representations (**ICLR**), 2025 [10+ citations]

[5] **Yuhui Wang**, Andi Wang, Di Wang, Dong Wang, "Deep Learning-Based Sensor Selection for Multimodal Recognition and Prognostics Under Time-Varying Operating Conditions", IEEE Transactions on Automation Science and Engineering (**IEEE TASE**), 2025 [10+ citations]